

AN APPLICATION OF GIS AND TIME SERIES ANALYSIS TO INVESTIGATE THE IMPACTS OF COVID-19 ON GROUNDWATER TREND FLUCTUATIONS CAUSED BY MANUFACTURING SITES AND RESIDENTIAL LIVING ON LONG ISLAND, NY



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Abstract

Long Island, NY heavily relies on groundwater supply for daily usage. The recent COVID-19 pandemic has significantly impacted daily living operations, including groundwater usage. As most citizens stayed home during the pandemic, we hypothesized that the amount of groundwater used would be reflected in the groundwater level fluctuations, which would positively correlate with the mean income level across different areas. We also hypothesized that there would be a positive correlation between manufacturing activity and groundwater level. To investigate our hypotheses, we collected groundwater data from 58 USGS groundwater monitoring stations on Long Island, then used ArcGIS software and R programming language to collect and process data from the US Census Bureau. We then performed a spatial correlation by using GIS methods and applied a linear regression on the groundwater level data to determine the trend and slope of each interval of stages of COVID-19. Our findings revealed no significant linear trends that correlated groundwater supply rates with mean income, employment rates, family or married households. However, we did find an inverse relationship between the number of manufacturing sites and groundwater slope trends during the pandemic. This can be interpreted as an increased groundwater usage rate in areas with a higher number of manufacturing sites.

Introduction

- Groundwater levels across the US are decreasing. Long Island is reliant on 4 aquifers: Upper Glacial, Magothy, Raritan Clay, and Lloyd Aquifers. In March 2020, covid-19 forced the world into a lockdown that lasted over a year. During the lockdown, the CDC recommended frequent handwashing as well as washing products like groceries. A 2022 study (Nemati & Tran 2022) has shown that due to covid, water usage has gone up significantly, due to changes in hygiene practices in direct correlation to the pandemic and the instituted stay-at-home orders.
- Hypothesis:** Because of covid-19 and its effects on water use, we hypothesize that a positive correlation between the amount of groundwater used and the average income level in a given area may be present. We also predict a significant correlation between groundwater usage and manufacturing within Long Island. (see Figure 1)

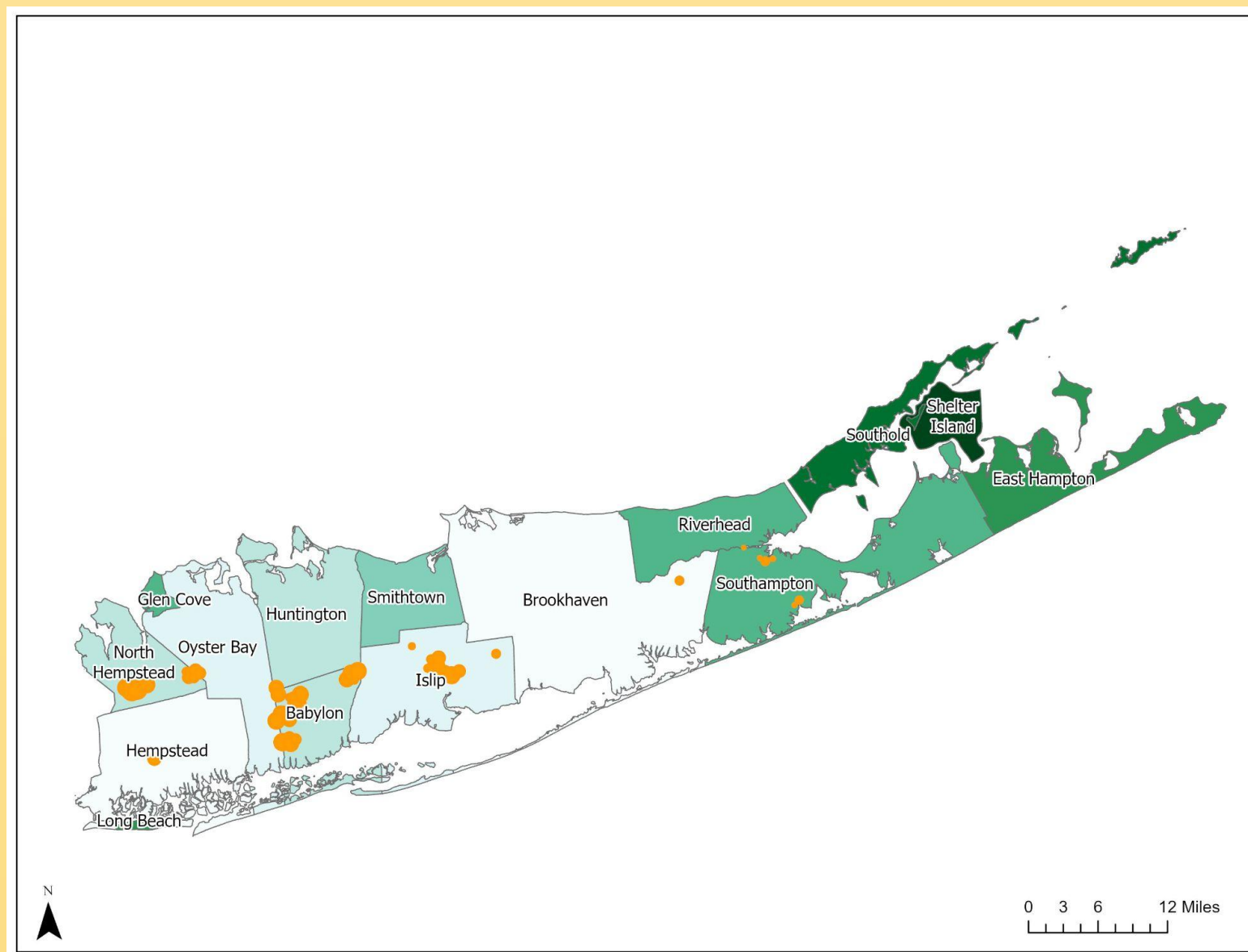


Figure 1: A GIS map of Long Island showing town location, mean income (green polygons) and manufacturing sites (orange polygons). The area and amount of manufacturers on the Island is shown on this map. The sites projected have 50 or more manufacturers.

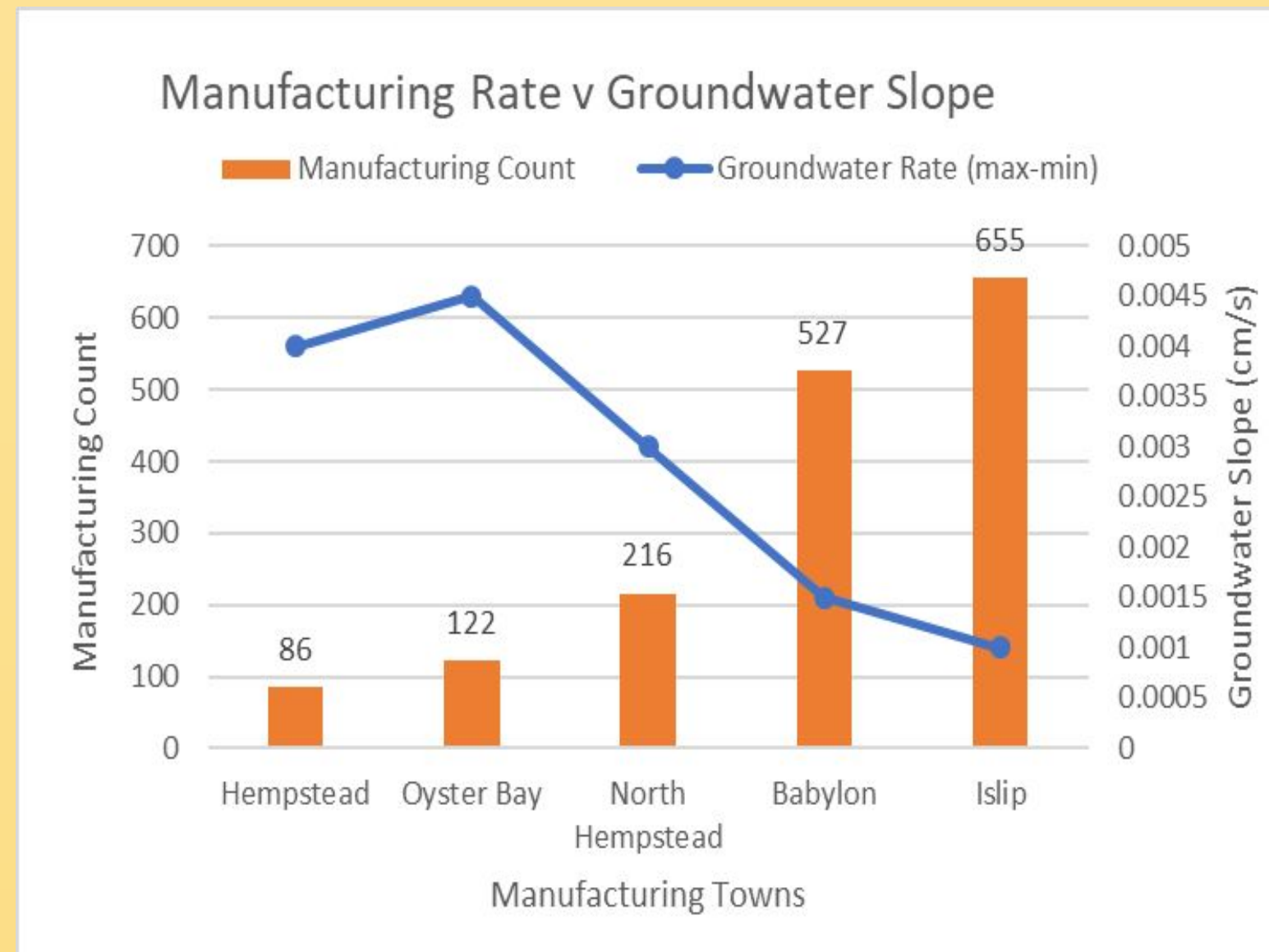


Figure 2: A graph showing the number of manufacturing sites, towns on Long Island, groundwater slope, and the rate of groundwater. We observe an inverse relationship between variables.

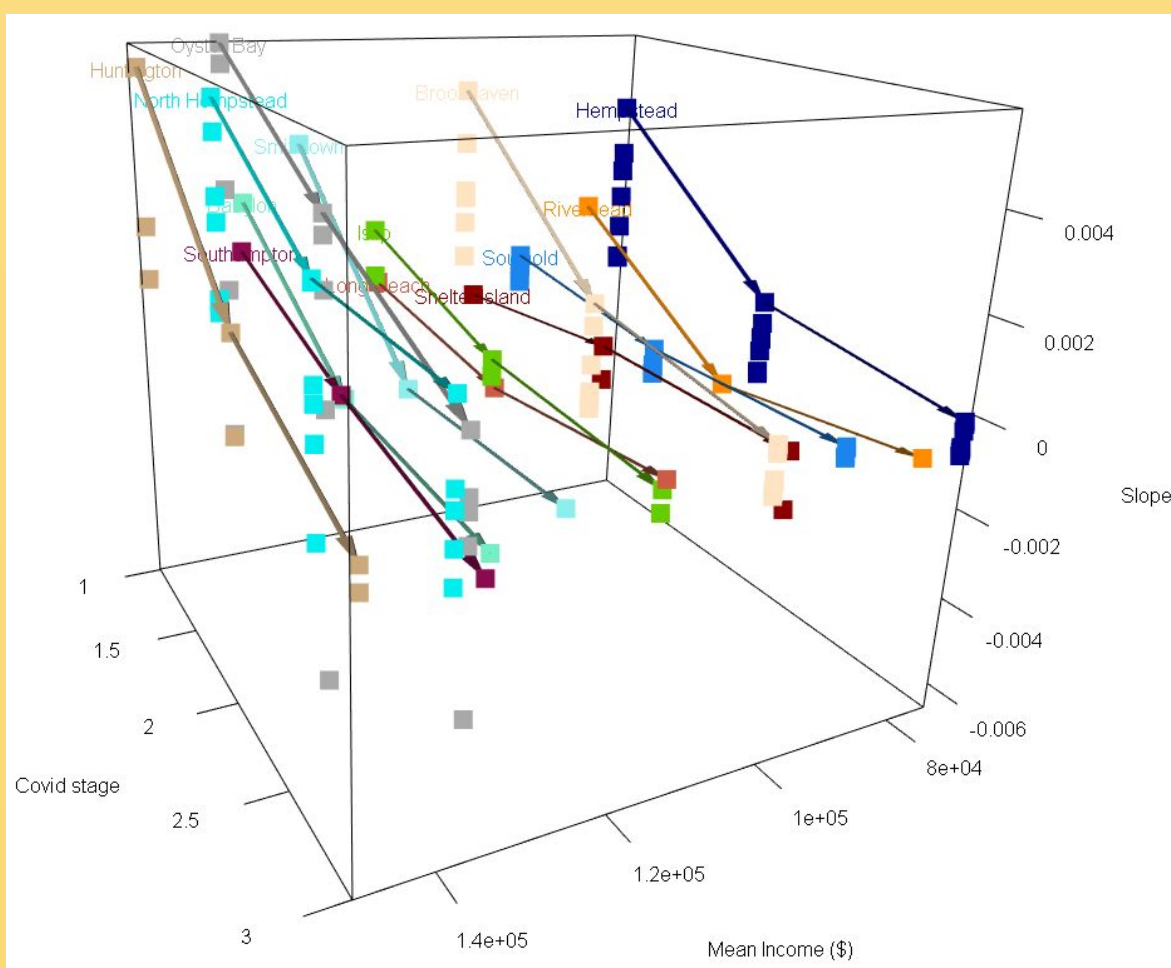


Figure 5: A three-dimensional graph showing towns, mean income, covid stage, and groundwater level trends (slope; determined as differencing with previous day). We observe a slight increase of variation in the data points as mean income increases.

Table 1: Amount of manufacturing sites per town in Long Island, groundwater slope maximum and the minimum, and the difference between max. and min. values of groundwater level trends at different covid time intervals.

	manufacturing count	slope (max)	slope (min)	difference
Oyster Bay	low	0.0060	0.0015	0.0045
Hempstead	low	0.0040	0	0.0040
North Hempstead	mid	0.0050	0.0020	0.0030
Babylon	high	0.0020	0.0005	0.0015
Islip	high	0.0010	0	0.0010

Results

- Using Statistical Analysis an inverse relationship was found between the number of manufacturing sites and groundwater slope rate. (Table 1 and Figure 2)
- Statistical Analyses showed no significant correlation between groundwater usage pre-, during-, and post-covid with mean income, employment rate, married households, or family households. (Figures 3 and 4)
- Slopes per interval were determined and provided p-values that indicated that predictor's changes are related to changes in the response value (rejects null hypothesis)
- There are no visible trend lines that would correlate a positive or negative relationship between income level and groundwater usage. (Figures 3 and 4)
- During Covid recharge rate rose in less populated areas; East Hampton, Hempstead, Huntington, Oyster Bay, Shelter Island, and Southold. (Figure 5)

Methods

- A spatial correlation was conducting using GIS methods of clipping the population and income polygons with groundwater trend levels. The GIS applications took place in both ArcMap and Rstudio.
- In Rstudio we used the following packages:
 - Sf
 - Cartography
 - MapsF
 - Celestial
 - Leaflet
 - Kza
 - Dplyr
 - Lubridate
 - BBmisc
- Demographic population data was retrieved from the US Census Bureau
- Groundwater level data was obtained from USGS
- The shapefiles came from the New York State Clearinghouse
- Distinction between high- and low-income areas was determined by mean income and employment rate gathered from the US Census
- To avoid seasonal discrepancies, the study was completed over three different time periods of moving average groundwater level data:
 - Pre-covid
 - 2018 to 2020-03-01
 - Covid
 - 2020-03-01 to 2021-06-01
 - Post-covid
 - 2021-06-01 to 2023-02-01
- Trend was determined by applying a linear regression on each time interval and on the groundwater level data that were averaged using the KZ filter
- Fit line was obtained and the associated slope/trend was determined using the p-values
- To calculate the manufacturing groundwater use:
 - Number of manufacturers in Long Island was counted
 - Groundwater slope was determined for each active station
 - Interpreted trend of the groundwater level was determined by subtracting the previous day after the application of the KZ filter
- The calculated number was considered the recharge rate

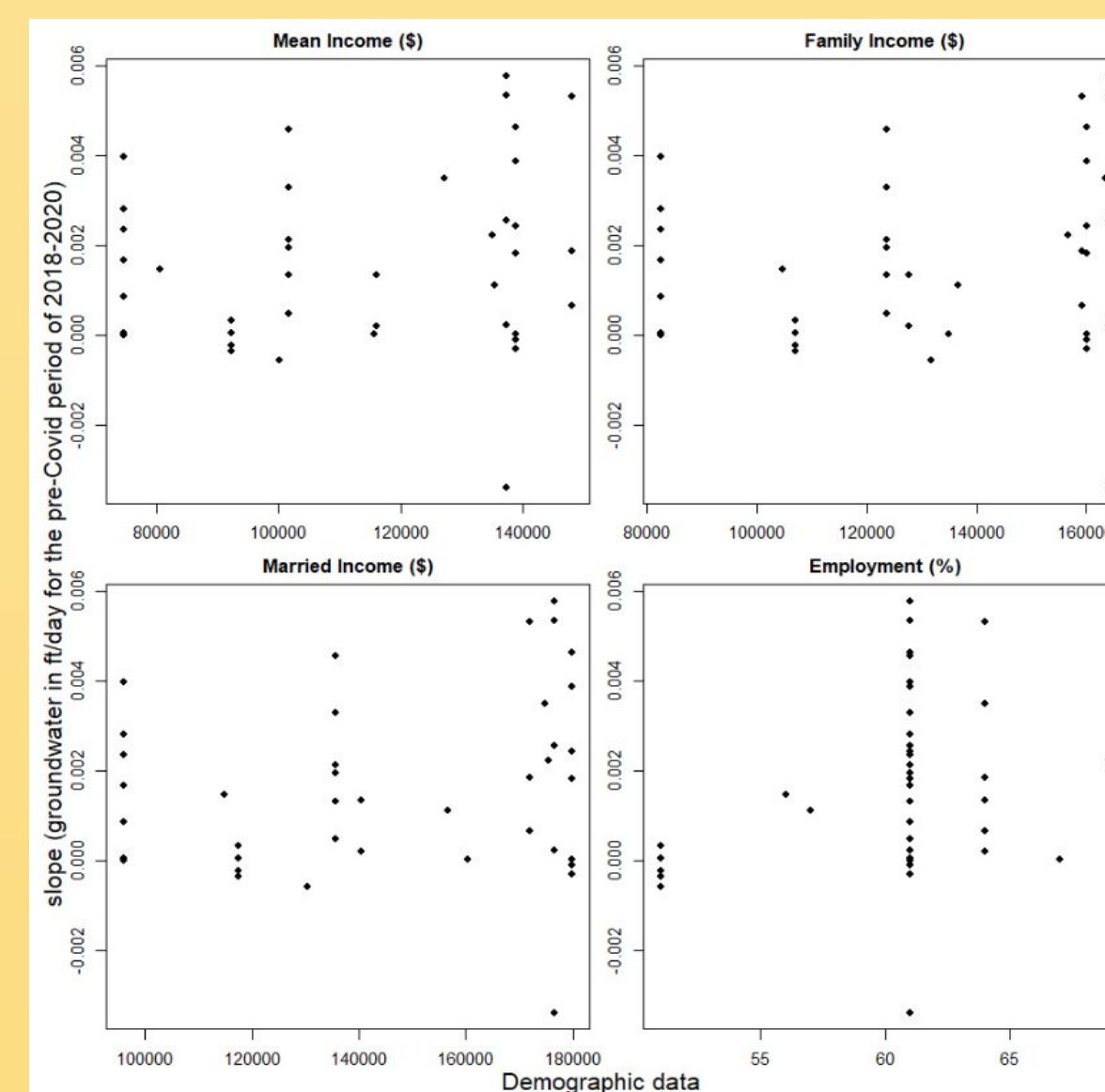


Figure 3: From left to right – Scatterplots showing trends of groundwater levels (slope) for each of the 58 USGS wells for the pre-covid period of 2018-2020 vs. mean income, vs. family homes, vs. married households, and vs. employment rate accordingly. There are no significant linear trends that correlate groundwater level trends (interpreted as a proxy for water usage) with these variables.

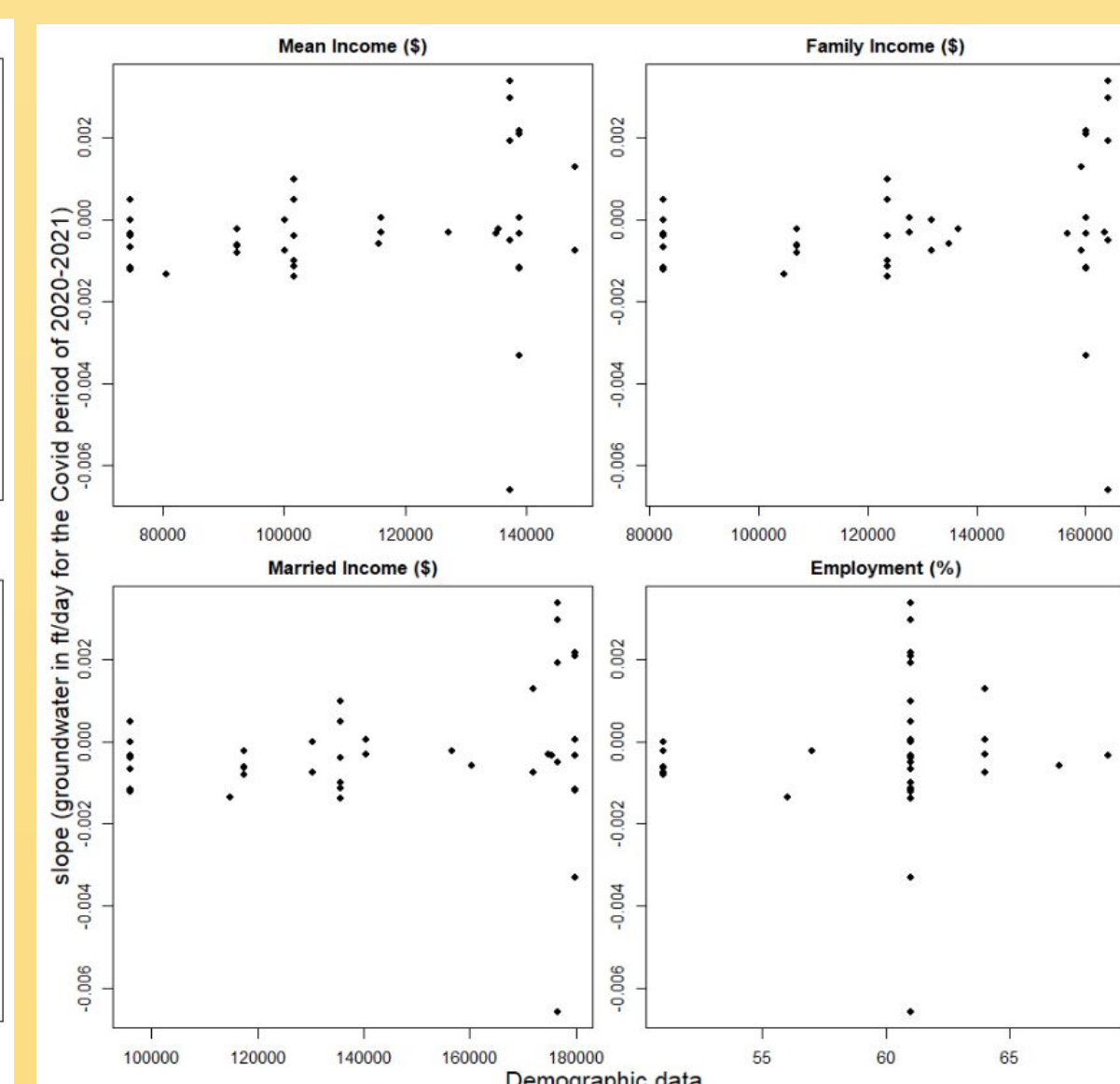


Figure 4: From left to right – Scatterplots showing trends of groundwater levels (slope) for each of the 58 USGS wells for the covid period of 2020-2021 vs. mean income, vs. family homes, vs. married households, and vs. employment rate, accordingly. There are no significant linear trends that correlate groundwater level trends (interpreted as a proxy for water usage) with these variables.

Discussion/Conclusion

- In Figure 5 over the course of the pandemic period, this graph shows that in higher income towns, the groundwater recharge slope has a higher variance than the groundwater recharge slope in the lower income towns. Could be that lower income areas tend to be more frugal with use of utilities.
- However, this slight variation (Figure 5) is not significant enough to say with certainty that there is a positive correlation between higher income and groundwater usage.
- Based on this study, residential activity had no significant effect on groundwater usage during the stages of the covid pandemic.
- Based on this study, the number of manufacturing sites and groundwater level trends had an inverse relationship during the stages of the pandemic.
- This model can be replicated and applied to other US states and areas.

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Authorship Contribution Statement

Hope, J.: editing, formatting, R coding, writing - Abstract, Introduction, Methodology, Results, Discussion, Conclusion; Morrison-John, M.: R coding, formatting, reference citing, writing - Introduction, Methodology; van der Vegt, M.M.: R coding, data analysis, data curation, editing, writing - Introduction, Results; Varnum, G.: reference citing, mapping, formatting, data curation, data analysis, writing - Introduction, Methodology, Results; Marsellos, A.E.: supervision, R coding, guidance, editing; Tsakiri, K.G. - R coding/providing KZ-Filter code for data processing.